Humanitarian Image and Text Classification with Gated Fusion

Dataset Proposal for Assignment 3 Milestone 1

Ho Chi Minh City University of Technology - VNU-HCM
Faculty of Computer Science and Engineering
Deep Learning and Its Applications | CO3133 | Semester 261

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Instructor: Lê Thành Sách | Prepared: 4 November 2026
M1 deadline: 18 November 2026, 23:59 GMT+7. M1 carries 15% of Assignment 3 [1].

Submission basis. The group reports that the instructor has waived prior dataset approval for this group. We therefore submit this M1 record without claiming a formal approval decision. No models have been trained for this proposal.

1 Research problem and objectives

We will classify a tweet and its attached image into one of five humanitarian categories. The prediction unit is one image-text pair; a tweet with multiple images contributes multiple records. CrisisMMD supports research on disaster information, but its social media content and historical event mix limit deployment claims [2,3].

Research question. Under fixed frozen encoders and leakage-controlled splits, does learned gated fusion improve macro-F1 over concatenation and unimodal models, and which modality interactions explain any gains or failures?

Our objectives are to establish four fair baselines and models, isolate the contribution of learned gating, and explain errors through paired examples and measured compute cost. Better multimodal performance is a hypothesis, not a promised outcome [1].

2 Dataset identity and provenance

The base resource is CrisisMMD v2.0 from QCRI. We use a custom five-class derivative of the agreed-label humanitarian benchmark, obtained through the third-party pranitchawla/crisismmd-mteb mirror, rather than the full eight-class QCRI configuration [4,5]. The mirror has 8,079 audited pairs; filtered_original retains 7,938. Annotations are one five-way categorical target per pair, with modality labels and confidence fields available in primary records.

Upstream license: CC BY-NC-SA 4.0 [4]. We will retain attribution and license notices, identify filtering changes, and observe the applicable noncommercial and share-alike conditions. No additional rights over source content are inferred.

Pinned dataset revisions and exact file hashes are recorded in the accompanying audit and remediation packages. The original agreed-label ZIP returned HTTP 403 during validation; archive equivalence remains unverified. The prior 245/245 image-byte matches prove only the checked subset, not archive equivalence.

3 Verified paired samples and preliminary EDA

All numbers in this section come from the completed validation/remediation evidence. No new data audit or model experiment was run. The primary set has 6,001 training, 986 development and 951 test pairs, 7,103 distinct tweet IDs and 7,071 connected groups. All five classes remain represented.

Table 1. Measured class counts for filtered_original. Source: remediation/tables/class_counts.csv; IDs match the locked manifests.

ID	Humanitarian class	Train	Dev	Test	Total
0	Affected individuals	71	9	9	89
1	Infrastructure and utility damage	601	80	81	762
2	Rescue volunteering or donation effort	901	148	126	1175
3	Other relevant information	1247	235	235	1717
4	Not humanitarian	3181	514	500	4195
	Total	6001	986	951	7938

Label mapping. Injured or dead people and missing or found people merge into affected individuals; vehicle damage merges into infrastructure and utility damage. Other categories keep their meanings. Retained records pass the primary per-modality agreement check before these merges; no target was relabelled during remediation.

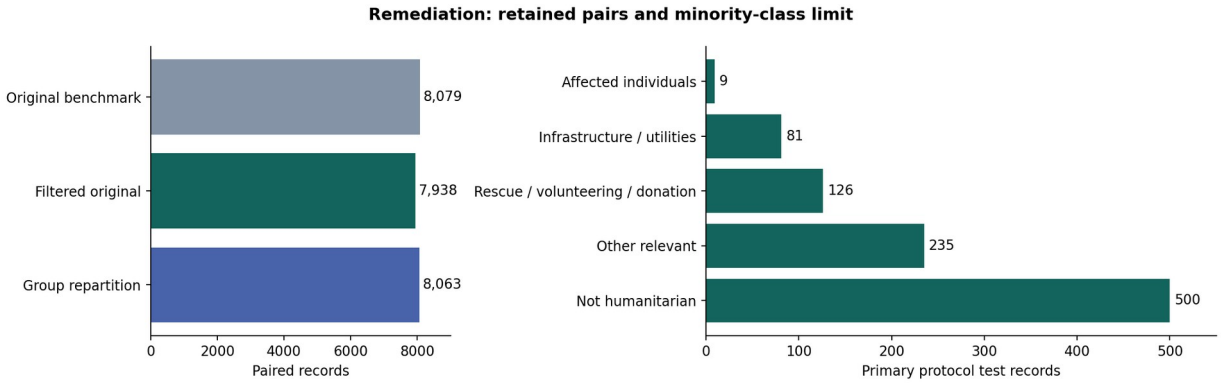


Figure 1. Measured retention and primary test-class counts, reproduced unchanged from the remediation report. The original 8,079-pair pool and filtered 7,938-pair set are distinct populations.

Table 2. Preliminary input statistics measured on the original 8,079-pair pool, not recomputed for the filtered set. Word counts use whitespace tokenization; they are not DistilBERT subtoken counts.

Original split	Median words	95th percentile	Median width / height
Train 6,126	14	20	900 / 675 px
Dev 998	14	20	869.5 / 641 px
Test 955	14	20	864 / 635 px

The original audit decoded all 8,079 images: file-level missing/corrupt-image and empty-text rates were 0%; primary joins and ID/path checks also passed. Retained rows inherit these checks. Decodable placeholders were separately excluded. The original pool has 652 heuristic encoding flags and 24 images with a side below 64 pixels; these are flags, not confirmed semantic errors. Their filtered counts have not been recomputed.

4 Alignment exclusions and evaluation protocol

Alignment links each image ID and path to its tweet ID, text and event metadata in pinned QCRI records. These are attached-image pairs, not random combinations of unrelated datasets [4,5]. Provenance joins establish record linkage, not the truth of every pictured location or claim.

Existing visual review found compatible examples: record 904472503850082305_0 shows a flooded roadway with a Harvey flood-impact tweet; record 909878988407033857_1 shows animal transport crates with a tweet about moving shelter dogs. Both remain in training. These observations illustrate compatibility and do not establish a dataset-wide semantic accuracy rate.

Table 3. Exact exclusions from the audited 8,079-pair mirror. Exclusions are disjoint; 8,079 - 141 = 7,938 retained pairs.

Reason	Train	Dev	Test	Total
Primary annotation discrepancy	7	0	0	7
Verified missing-image placeholder	4	1	4	9
Duplicate or leakage-related component member	114	11	0	125
Total	125	12	4	141

Annotation uncertainty. All seven quarantined training targets are not_humanitarian. QCRI event TSVs label their text other_relevant_information and their image not_humanitarian; QCRI humanitarian JSON and mirror targets agree with the image label. IDs, text and paths match. The cause may concern source versions or derivation rules and remains unresolved. We preserve the original evidence and labels; confidence scores do not justify silently relabelling these rows.

Leakage prevention. We apply the frozen remediation rules before any training:

1. Build transitive components from shared tweet IDs, byte-image hashes, decoded-pixel hashes, raw-text hashes and normalized-text hashes, plus 81 visually confirmed duplicate or shared-asset candidate edges.
2. Use the audited normalization: HTML unescape, Unicode NFKC, casefold, remove URLs, leading retweet mention prefixes and mentions, then join Unicode word tokens. Identical cleaned input is grouped conservatively even when source posts differ.
3. After forming groups, quarantine the 7 discrepant rows and 9 confirmed placeholder rows. For a component spanning eligible splits, retain only its original test rows, otherwise dev rows, otherwise train rows. Exclude lower-priority members; move no retained record.
4. Lock the primary manifests, including identity, label and content-hash fields. Primary filtering is deterministic and uses no random seed. Keep whole tweet/component groups within one split.

Remediation traced all 17 exact-image cross-split groups (54 records) and reviewed 134 pHash candidates: 70 duplicates, 11 shared-asset variants, 38 distinct observations and 15 placeholder edges. Similar weather templates from different advisories were not treated as duplicates. The 51 checks across both candidate protocols and independent manifest verification passed; 26 result-file hashes matched a second run.

Protocol limits. Test is a custom filtered set of 951 pairs, not the untouched benchmark; published scores are contextual, not directly comparable. Events overlap across splits, so this evaluates the source event mix rather than unseen-disaster generalization. Unknown semantic or heavily edited duplicates can escape the declared checks. The 8,063-pair group repartition is backup only and will not be selected using model scores.

5 Planned models and training procedure

All following methods are plans for M2-M3. We will use TorchVision ResNet-18 with IMAGENET1K_V1 weights and the Hugging Face distilbert/distilbert-base-uncased encoder [6,7]. Both

encoders remain frozen in the primary comparison. Library components provide pretrained encoders and tokenization; we will implement the projections, classification heads, fusion and experiment runner.

Table 4. Four-model roadmap. Each model predicts the same five classes on identical primary splits; no result is claimed.

Model	Representation and operation	Milestone
Image only	512-dimensional pooled image feature, 128-dimensional projection, classifier.	M2
Text only	768-dimensional masked-mean token feature, 128-dimensional projection, classifier.	M2
Concatenation fusion	Concatenate the two 128-dimensional projected vectors; 128-unit hidden classifier.	M2
Gated fusion	Scalar sigmoid gate from the concatenated projections; weighted sum of image and text vectors; 128-unit hidden classifier.	M3

The improved design is intermediate feature fusion inspired by gated multimodal units [8]. The image receives weight g and text receives weight 1 - g, with g learned per pair. A gate value is a model response, not proof of causal modality reliability. All heads end in five logits; projections and classifier widths are fixed in advance.

5 1 Preprocessing and alignment

Images will be decoded as RGB and transformed with the selected ResNet weights recipe: resize short side to 256, center crop to 224, rescale and apply its ImageNet normalization [6]. This deterministic primary recipe avoids stochastic augmentation when caching frozen features. Text will use the audited normalization, then the uncased tokenizer with a proposed limit of 128 subtokens, attention masks and padding. Actual truncation and unknown-token rates have not yet been measured.

Expected tensors are image B x 3 x 224 x 224; token IDs and masks B x 128; targets B; logits B x 5. These are specifications, not an executed example batch. Cache and data-loader rows will be keyed by immutable row_id/image_id, with label/group assertions before feature joins.

Both modalities are required for the fusion benchmark. If an expected file or feature is unavailable later, the loader will stop and log the failure rather than substitute a different pair or silently alter a split. Missing-modality deployment robustness is outside the validated scope.

5 2 Optimization and model selection

Initial head-training settings are Adam, learning rate 0.001, weight decay 0.0001, batch size 64, dropout 0.2 and at most 30 epochs. Early stopping uses development macro-F1 with patience 5; ties use lower development loss, then the earlier epoch. These settings will be versioned if development-only tuning changes them.

Image-only and text-only losses are class-weighted cross-entropy; fusion uses the same supervised loss on the paired target, with no auxiliary modality losses. Weights are derived only from the 6,001 training labels using inverse class frequency and held fixed across models. Encoder fine-tuning is not planned for the required comparison; only projections, fusion and heads are optimized.

Use training seeds 42, 43 and 44 for each model, shared preprocessing/cache versions and the same checkpoint rule. The seed values are planned and have not produced any training runs. Test predictions will be generated only after configurations and development-selected checkpoints are frozen.

6 Evaluation and controlled analysis

Primary metric: macro-F1 across all five fixed class IDs. Secondary outputs: accuracy, per-class precision/recall/F1, class support and raw plus row-normalized confusion matrices [1]. Accuracy alone is inadequate because not humanitarian dominates the set. Report mean and standard deviation over the three seeds; show individual-run results and total/trainable parameters.

The affected-individual class has only 9 dev and 9 test pairs. One changed test decision shifts its recall by 11.1 percentage points. Report this limitation explicitly and use paired bootstrap intervals resampling test connected groups when quantifying model differences. Small-class results remain exploratory even if the point estimate improves.

Table 5. Planned comparisons and decision criteria. Fixed controls are the primary manifests, pretrained weights, feature recipe, loss, optimizer budget, seeds and development checkpoint rule.

Comparison and hypothesis	Changed factor	Evidence and conclusion criterion
Image versus text The modalities have different error patterns.	Available modality.	Macro-F1, per-class F1 and paired error sets; identify the stronger unimodal model using dev only.
Concatenation and gating versus unimodal models Joint representations may recover complementary evidence.	Fusion method or modality combination.	Report test macro-F1 differences and intervals against concatenation and the dev-selected unimodal reference; allow no gain or inconclusive results.
Gated versus fixed gate Learning the gate may help beyond uniform averaging.	Set gate output to 0.5; retain the same encoders, projections, classifier and training settings.	A controlled gate ablation with macro-F1, per-class errors and cost. Report the trainable-parameter difference; do not attribute every effect solely to attention or trust.

6 1 Modality interaction and error analysis

We will cross-tabulate correct and incorrect predictions from both unimodal models and each fusion model. Support cases include both unimodal predictions agreeing correctly; complementarity includes one correct unimodal prediction recovered by fusion. Operational conflict means the unimodal predictions disagree and is then assessed by paired visual/text review. Fusion harm means fusion is wrong when at least one unimodal model is correct.

Because this is an agreed-label benchmark, we will not treat source annotation disagreement as ground-truth modality conflict. Gate distributions may illustrate behaviour but will not establish explanations on their own. Representative successes, failures and conflict cases will show text, image, target and predictions. Error groups will include weak/irrelevant visuals, noisy text, ambiguous labels, minority-class confusion and event-specific shortcuts; frequencies will be measured when results exist.

The earlier data audit inspected a small set of train/test pairs for quality. No model predictions or performance-based choices were involved. From this point, development data controls model and hyperparameter selection; test will not be used to select a seed, protocol, subset, encoder or fusion design.

6 2 Compute cost reporting

For each main configuration, record hardware, RAM/VRAM, software/model revisions, preprocessing and frozen extraction time, head-training wall time, epochs, peak memory and inference latency/throughput under stated batch size and warm-up conditions. Separate one-time feature extraction from head fitting and total end-to-end inference so caching does not hide encoder cost.

7 Resources reproducibility and risk controls

Table 6. Measured storage and provisional resource estimates. Estimates are planning allowances, not observed runtimes or confirmed hardware access.

Item	Evidence or estimate
Existing mirror files	Measured: four pinned Parquet shards total 860,962,496 bytes, approximately 821 MiB. Images and tweet text are already cached.
Frozen feature cache	Estimated float32 feature payload: 7,938 x (512 + 768) x 4 = 40,642,560 bytes, approximately 38.8 MiB, before IDs and metadata. No features have been extracted.
Working storage	Reserve approximately 3-5 GB for data, downloaded pretrained weights, features, head checkpoints and logs, excluding the environment and temporary downloads.
Compute plan	No hardware allocation is verified. Target one 8 GB GPU if available for extraction, with CPU extraction/head fitting as fallback. Provisionally reserve 1-3 GPU-hours plus 1-4 CPU-hours for extraction and 15 head fits (four models plus gate ablation, three seeds). Validate the allowance on 100 training pairs after M1 and revise from measured throughput.

No additional training subset is planned: all 6,001 train pairs and complete dev/test manifests are used for reported experiments. A development-only smoke check may use at most 256 training records selected deterministically with seed 42, preserving whole groups and class coverage; its IDs must be logged and its scores must not be reported as main results. Resource limits will first reduce batch size or use cached/CPU features, not alter the primary split.

Reproducibility record. Use the locked train.csv, dev.csv and test.csv in A3-M1/remediation/manifests/filtered_original, with hashes in remediation/SHA256SUMS.txt. Version: crisismmd5-mirror-remediation-20261104-v1. Preserve row IDs, labels, original memberships, groups, exclusions, source paths and hashes. For M2-M3, save configs, seeds, encoder/tokenizer and package versions, run IDs, logs, checkpoints and commit/tag links. Existing validation/remediation scripts and pinned sources remain unchanged.

Table 7. Principal risks and mitigations.

Risk	Mitigation and limitation
Unverified archive equivalence and seven source-label discrepancies	Name the mirror-derived custom version; retain the 7-row quarantine and original evidence. Do not claim exact archive reproduction or silently substitute the 8,109-row naive reconstruction.
Rare-class instability and correlated posts	Keep class/group support visible; use group-aware uncertainty, qualitative analysis and restrained claims. Two retained primary groups contain multiple labels; grouping does not authorize relabelling.
Encoding flags low-resolution images and shared templates	Retain immutable text; measure tokenizer behaviour in development. Do not infer codec fixes from heuristics. The 134 reviewed candidates distinguish actual reuse from distinct weather advisories; further semantic duplicates may remain.
Event bias and selected agreement labels	Limit claims to the historical 2017 social media event mix. Agreed-label selection underrepresents natural cross-modal disagreement and does not validate crisis-response deployment.

7 1 Milestone plan and submission scope

M1, 18 November: submit the dataset proposal and existing audit evidence. M2, 25 November: image-only, text-only and concatenation results. M3, 2 December: gated fusion, gate ablation,

interaction/error/cost analysis and the full report, slides, video, page and checkpoint/reproduction package. All deadlines are 23:59 GMT+7 [1].

Repository/Pages publication, AI log updates and LMS upload remain group submission actions; this preparation step makes none of those edits. LMS-specific naming, file formats and packaging must be checked on the course page. Prior approval is waived as reported by the group, not recorded as a formal approval.

8 AI Usage Disclosure

Generative AI assisted this M1 work. The record below identifies its contribution; students remain responsible for understanding and final verification [1, Section 5].

Required field	Disclosure
Tool and model/version	ChatGPT Work with the Codex assistant. Underlying model/build not independently verified. No training or model results.
Member who used the tool	Nguyễn Huỳnh Nam Quốc, 2453089 (requester-confirmed).
Time or stage	4 November 2026; M1 source review, reuse of audited evidence, planning and proposal preparation.
Purpose	Handbook/source checks, English drafting/editing, Word/PDF formatting and compliance audit.
Affected sections	Sections 1-9, Tables 1-7, reused Figure 1 and the compliance checklist.
Representative prompt	Prepare an English M1 proposal using verified EDA/manifests; disclose archive uncertainty; no training or GitHub edits.
Editing and verification	Assistant cross-checked claims against the handbook, locked CSV/JSON and primary references; rendered and inspected pages. Completed independent student review is not claimed.
Responsible final verifier	Nguyễn Huỳnh Nam Quốc, A3 lead; review, correct and verify the submission before upload.
Verification sources	Handbook [1]; pinned QCRI/mirror [4,5]; audit/remediation tables, manifests and checksums; references [2,3,6-8].

This disclosure and its prompt/verification record must also appear on the required landing page, assignment page and AI log. This step leaves existing logs unchanged.

9 References

[1] Lê Thành Sách. Course Project Handbook, CO3133, Semester 261, HCMUT, 2026. Part I and Sections 7.6, 24-32. PDF pp. 5-16 and 27-31. Course LMS

[2] F. Alam, F. Ofli and M. Imran. CrisisMMD: Multimodal Twitter Datasets from Natural Disasters. ICWSM, 2018. arXiv 1805.00713

[3] F. Ofli, F. Alam and M. Imran. Analysis of Social Media Data using Multimodal Deep Learning for Disaster Response. ISCRAM, 2020. arXiv 2004.11838

[4] QCRI. CrisisMMD v2.0 dataset card, primary annotations and CC BY-NC-SA 4.0 license. Pinned revision 10a5626ba112ab9c50369c473a2062bcf205d708. Primary resource

[5] pranitchawla. crisismmd-mteb third-party mirror. Pinned revision a935fda3cee8a5250746a8d1d6278214d615cc89. Mirror resource

[6] PyTorch TorchVision. ResNet-18 documentation and IMAGENET1K_V1 transforms. Official documentation

[7] Hugging Face DistilBERT. distilbert/distilbert-base-uncased model card and model documentation. Model card

[8] J. Arevalo, T. Solorio, M. Montes-y-Gómez and F. A. González. Gated Multimodal Units for Information Fusion, 2017. arXiv 1702.01992